

CYCLESense ANALYTICS

Forecasting solutions for urban bike-sharing systems

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1 Executive Summary

Business Context CycleSense Analytics builds demand forecasting solutions for urban bike-sharing systems. For our client Capital Bikeshare in Washington D.C., getting an accurate 14-day forecast is important because it affects everyday operations like Station rebalancing, Maintenance scheduling, Workforce planning, and Managing budgets. If demand is underestimated, stations can run empty and revenue is lost. If it is overestimated, time and labor are wasted moving bikes unnecessarily. Either way, it impacts rider experience and makes the overall system less efficient.

Key Findings The data shows strong weekly seasonality and a clear temperature effect ($r = 0.63$). Among the models we tested, Damped Holt-Winters with multiplicative weekly seasonality worked best and is considered as the primary forecasting tool. On a 14-day test set, it gave an MAE of 818 rentals and sMAPE of 16.2%, which is around 29% better than a seasonal naive baseline. We also tested a SARIMA(1,1,1)(1,1,1) with weekly seasonality as an alternative (MAE 843), and a simple 50/50 average of the two models (MAE 824). For uncertainty, bootstrapped 95% prediction intervals gave 85.7% empirical coverage with an average width of about 3,733 rentals. This is narrower than SARIMA's analytical intervals (about 4,144) and scores better on MSIS (42.6 vs. 47.3).

High-Level Implications For routine day-to-day operations, the forecasts are accurate enough to guide fleet planning, but there are a few gaps. First, Federal holidays and the Christmas/New Year window, where demand collapses 40-60% below model expectations, and sudden weather changes like rain/snow also create short term errors that the model does not handle well. Our final recommendation is to use Damped Holt-Winters as the main forecasting model, add holiday adjustments, include short-term weather forecasts, and plan bike deployment using prediction intervals based on demand levels (maybe to use the 80th percentile prediction interval instead of the point forecast).

2 Business Context & Data Description

Company and Operational Setting CycleSense Analytics is responsible for forecasting that works with urban bike-share operators. The current engagement is with Capital Bikeshare, the public bike-share system serving Washington D.C. and surrounding counties. Over a one to two week planning period, Capital Bikeshare must coordinate four connected operational tasks (i) overnight rebalancing routes that move bikes between stations to match expected morning demand, (ii) staff and field-technician schedules, (iii) maintenance windows that are best placed on low-demand days to minimize rider impact, and (iv) operating-cost forecasts that feed weekly budget allocations. A 14-day horizon is important because operational decisions like truck routing and bike redistribution, and workforce scheduling are planned for one to two weeks ahead, not the same day.

Forecasting Objective The primary deliverable is a daily total rental forecast for the next 14 days, accompanied by 80% and 95% prediction intervals. The point forecasts, which estimate expected rentals and the prediction intervals, help measure uncertainty and show the expected demand range. The intervals are there for days when the cost of being wrong is not equal in both directions, for example, a summer weekday where empty stations lose riders, or a holiday week where too many deployed bikes just sit in the rain.

Dataset Description The dataset contains hourly bike rental counts from Washington D.C. covering two full years, January 1, 2011 through December 31, 2012. There are 17,379 hourly records in total, which we aggregated to 731 daily observations for forecasting. The data is clean: there are no missing days and no null values. In 76 out of 731 days, there are fewer than 24 hourly records, simply because there was no activity in the late-night hours. However, this does not affect the daily totals. Table 1 lists the variables we used.

Table 1: Dataset variables and their roles in the analysis

Variable	Description	Type	Role
cnt (target)	Total bike rentals = casual + registered users (this is what we forecast)	Integer	Target
temp / atemp	Temperature and feels-like temperature, normalized 0 to 1 (we multiply by 41°C)	Continuous	Driver
hum	Relative humidity, normalized 0 to 1 (we multiply by 100%)	Continuous	Driver
windspeed	Wind speed, normalized 0 to 1 (we multiply by 67 km/h)	Continuous	Driver
weathersit	1=Clear, 2=Mist/Cloud, 3=Light rain/snow, 4=Heavy rain	Ordinal	Driver
season	1=Spring, 2=Summer, 3=Fall, 4=Winter	Categorical	Calendar
holiday / workingday	1= weekday (not weekend or holiday), 0 otherwise	Binary	Calendar
weekday / hr	Day of week (0=Sun through 6=Sat); hour of day (0 to 23)	Categorical	Calendar

All forecasting is done at the daily level over a 14-day horizon, targeting the univariate series *cnt*. On average, the system sees about 4,504 rentals per day with a standard deviation of 1,937. The minimum was 22 (a heavy rain day) and the maximum was 8,714. The series is always positive, integer-valued, and shows strong weekly patterns and sensitivity to weather. We explore these features in the next section.

3 Exploratory Analysis & Insights

Before building any models, we carried out five different exploratory analyses to understand the structure of the data: (1) an overall time-series plot (2) a basic seasonal breakdown (3) patterns by hour of the day and day of the week (4) weather and seasonal effects and (5) ACF and PACF plots. These results directly guided our choice of forecasting methods in Section 5.

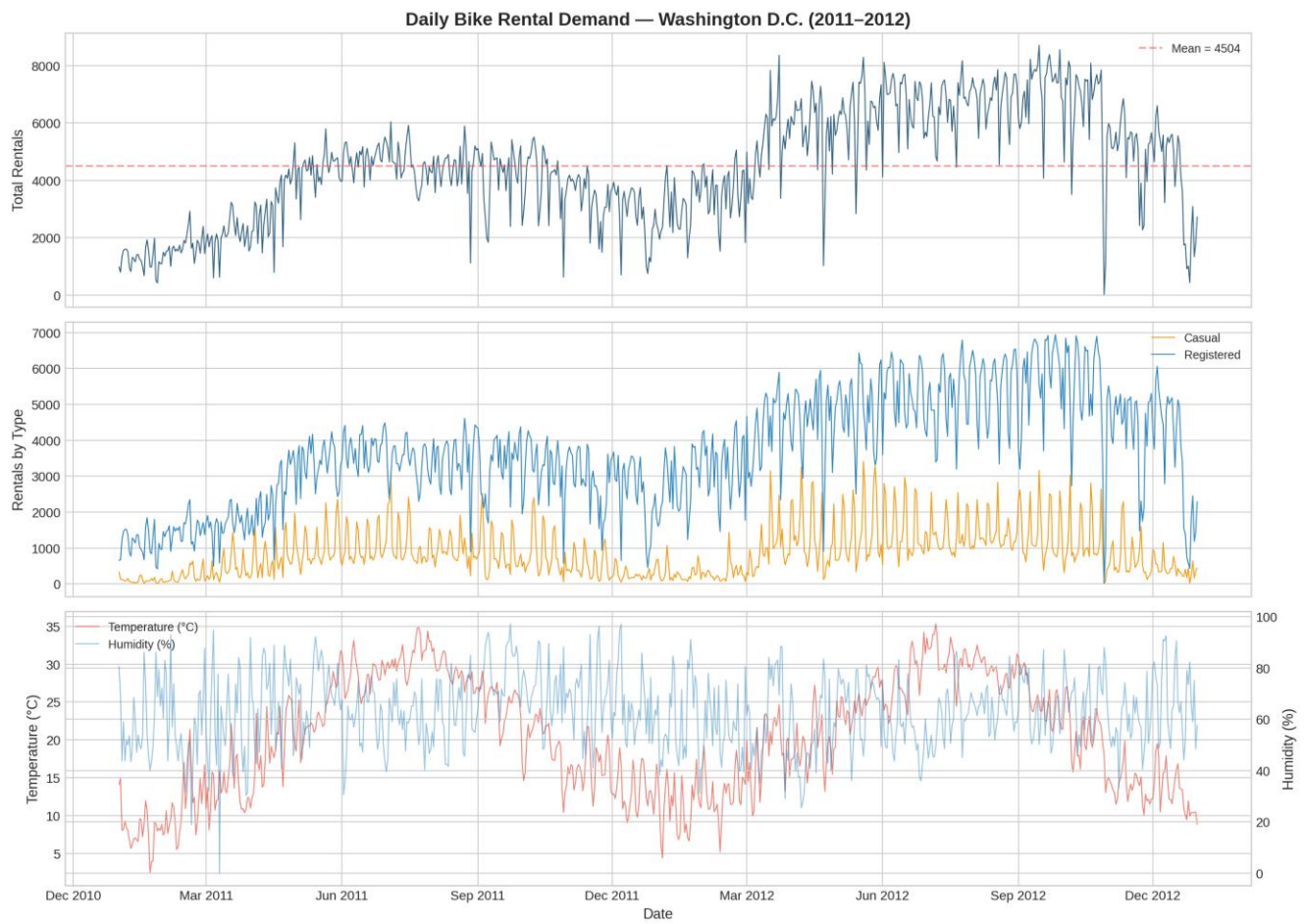


Figure 1: Daily rental counts, user type split, and weather variables (2011-2012)

Trend and growth Daily demand grew by roughly 65% from 2011 to 2012. Most of that growth came from registered users. Casual users are more active on weekends and more sensitive to weather. On an average day, registered users make up about 81% of all trips and casual users about 19%. This distinction is important for forecasting: registered demand follows predictable calendar patterns tied to commuter behavior, while casual demand is more driven by weather and adds most of the noise to our target series.

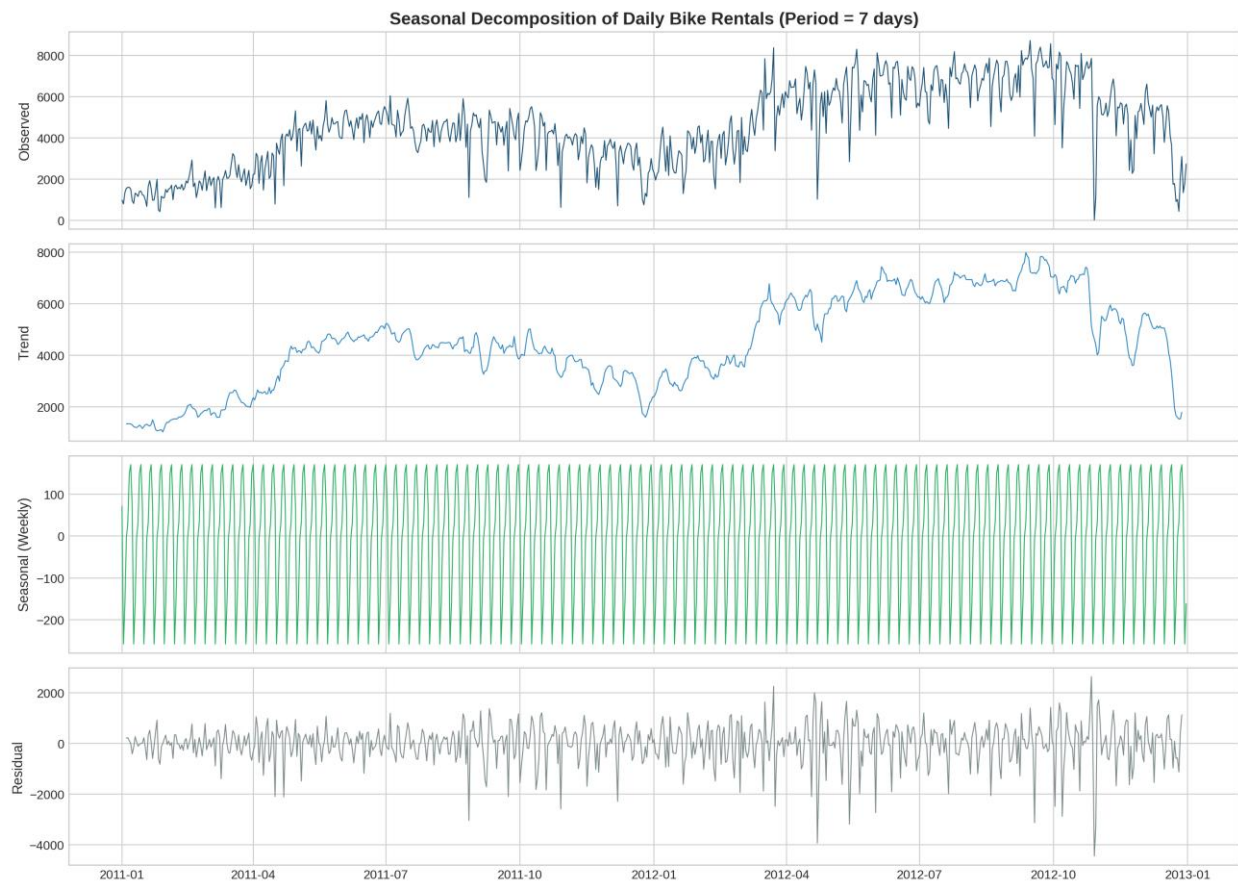


Figure 2: A clear upward trend through summer 2012 and steady weekly cycle

Seasonality Using a classic additive decomposition with a 7-day period (Figure 2) gives us a clean breakdown of three parts. The trend is smooth and upward. The weekly cycle is stable and adds or subtracts about 15% of the trend value. The residual is the remainder after removing trend and seasonal effects, like sudden weather changes and holiday effects. An interesting finding emerges when comparing user segments. Registered users peak on weekdays and decline on weekends, whereas casual users exhibit the opposite pattern. When aggregated, these two opposing patterns partially cancel each other out. Consequently, the weekly seasonality in total demand is weaker than the weekly pattern observed for either user group individually.

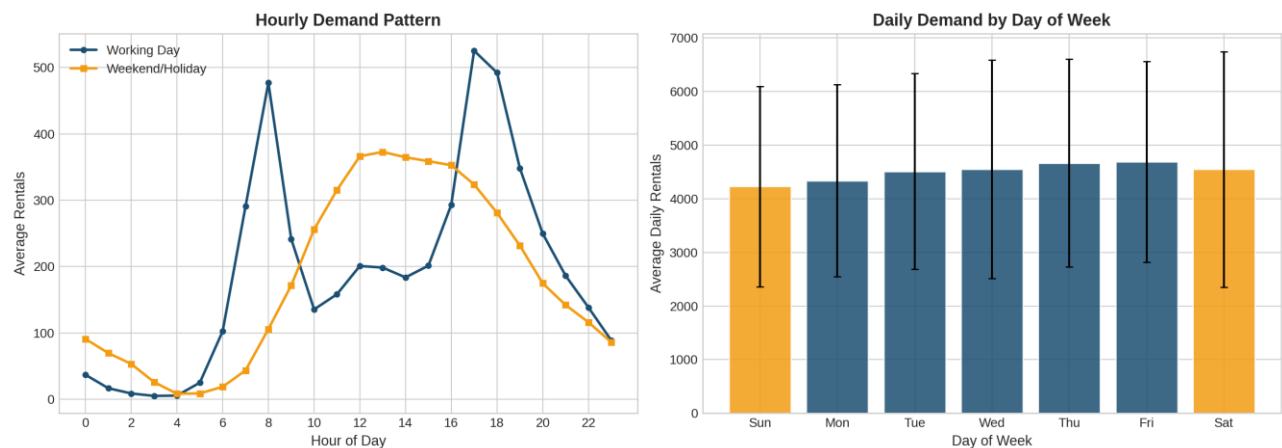


Figure 3: Hourly demand profiles and average daily rentals by day of week

Hourly and weekly patterns On working days, two sharp peaks are observed: one at 8 AM and another between 5 and 6 PM, reflecting usual commuter behavior. On weekends, there is only one main peak during midday around 1 to 3 PM, consistent with leisure riding. Examining daily totals by day of the week, working days average approximately 18% more rentals than weekends. Friday is the busiest weekday, while Sunday is the quietest day overall. These patterns motivated the use of multiplicative weekly seasonality in the Holt-Winters model.

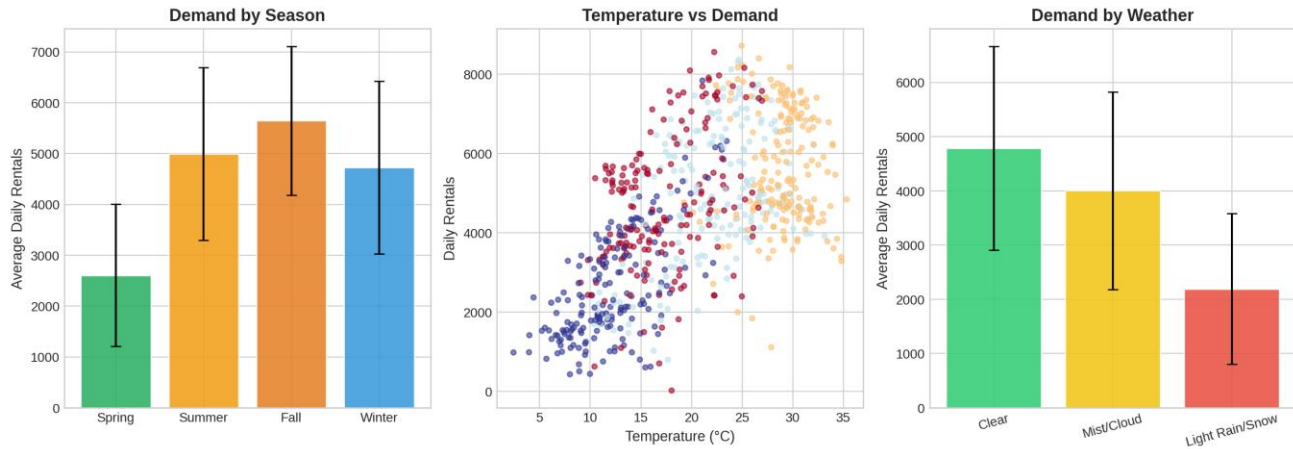


Figure 4: Demand by season (left), temperature vs. demand colored by season (center), and demand by weather category (right)

Weather and seasonal effects Temperature turned out to be the single strongest predictor of how many bikes get rented. The relationship is roughly linear up to about 30 degrees Celsius and then it plateaus. Rain matters a lot too: light rain or snow (weather category 3) cuts demand by about 40% compared to clear days, and heavy rain basically shuts things down. Of the four seasons, fall has the highest average demand at 5,372 rentals per day. This is likely because fall temperatures are comfortable and by that point in the year the registered user base is at its largest. There were three days of heavy rain (weather category 4) that had extremely low demand. We kept those in the data rather than removing them as outliers, since they represent real weather events that the system experienced.

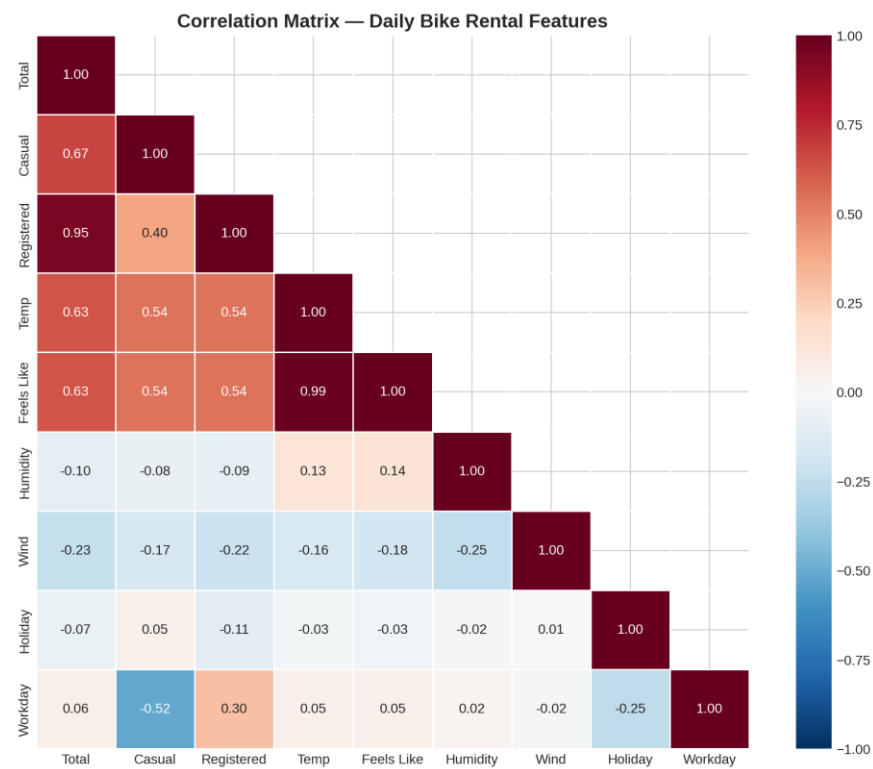


Figure 5: Pearson correlation matrix of daily features

Correlation structure Looking at the correlation matrix, temperature is clearly the main external driver, with a correlation of 0.63 with total daily demand. Humidity has a weak negative correlation (-0.10), and wind speed shows almost no correlation at all. The working day variable raises the overall level of demand but does not really change the shape of the hourly pattern. Since temperature is the only strong external predictor, we felt that starting with a univariate seasonal model made sense. We note in Section 8 that adding weather covariates through SARIMAX or machine learning is something we could improve in future work.

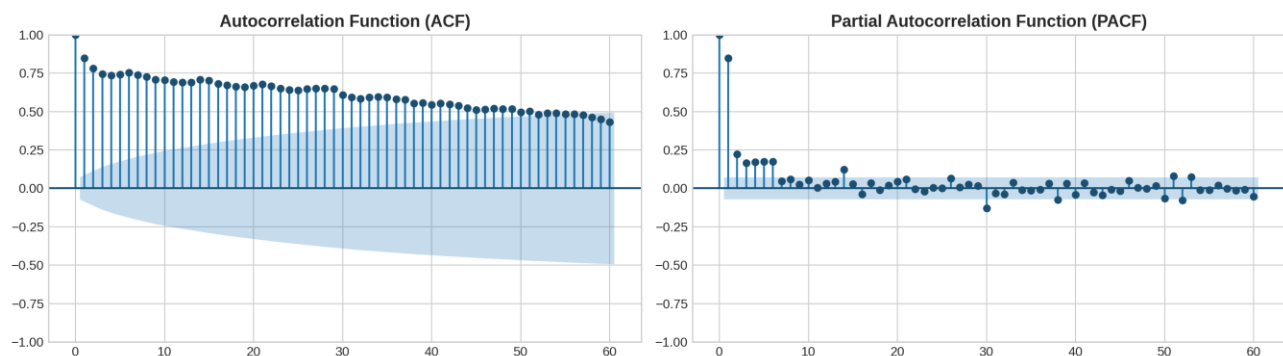


Figure 6: ACF and PACF of daily total rentals up to 60 lags

ACF and PACF The ACF decays slowly and shows clear spikes at lags 7, 14, and 21, which points to a weekly pattern along with a persistent level. The PACF drops off after lag 1, with another noticeable spike at lag 7. This pattern lines up with a $\text{SARIMA}(1,1,1)(1,1,1)_7$, and it also supports setting `seasonal_periods = 7` in the Holt-Winters model.

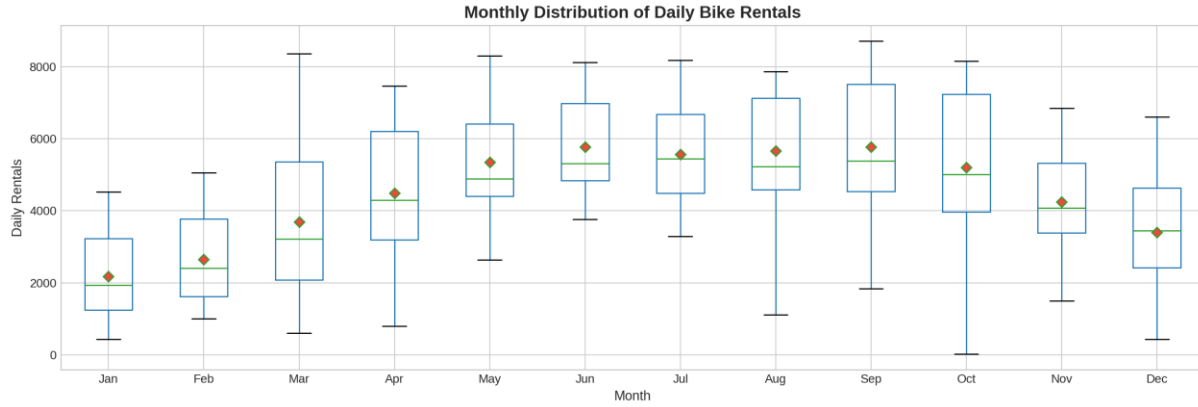


Figure 7: Monthly distribution of daily rentals (2011-2012 pooled). Demand peaks between May and October

Summary of EDA findings Overall, this series is well suited for seasonal exponential smoothing. It has a smooth trend, a stable weekly cycle, approximately normal residuals, and a strong but understandable dependence on temperature. Two risks are already visible at this stage. First, federal holiday weeks - especially the period from Christmas to New Year cause sharp drops in demand that a seasonal model cannot anticipate. Second, sudden weather changes can disrupt short-term forecasts. We return to both issues in the risk analysis section.

4 Data Preparation

Temporal aggregation The raw data comes to us on an hourly basis, but the decision making that we are helping with (route balancing and weekly scheduling) occurs at the daily level. Therefore, we aggregated the hourly rentals to give us daily totals and averaged the temperature, humidity and windspeed to get daily averages.

De-normalization The weather data in the raw set are normalized between 0 and 1, which works fine for modeling purposes, but is difficult to use in explanation. So, we denormalized these values so that they are all to be interpreted in a physical unit. We multiplied temperature by 41 to get Celsius, humidity by 100 to get percent, and wind speed by 67 to get km/h. After denormalization, all plots, tables, and residuals show directly interpretable results.

Missing data and outliers There were no missing days and no gaps in the time series data, so we did not need to fill in any missing information. The three days that had category 4 weather (heavy rain) had relatively low numbers of rentals (< 100) and they were included in our test set since this was a weather induced phenomenon rather than something that needed to be filtered out. Outlier analysis was not performed for daily values.

Train/test split We used two evaluation setups. For the primary test, we held out the last 14 days of the series (December 18 to 31, 2012) since that matches the operational forecast horizon our client plans around. On top of that, we ran rolling-origin cross-validation with 10 windows of the same length, stepping backward two weeks at a time, with a minimum of 365 training days per window. The holdout gives us one clean scorecard. The rolling CV lets us see how stable the errors are across different parts of the year.

Reason behind these choices We aggregate the data to daily level to match how planning decisions are made. This removes hourly detail, but that tradeoff is acceptable given the use case. The heavy rain days were kept since dropping them would make the model too confident under bad weather conditions. For evaluation, we used rolling origin cross validation instead of a

random split because demand increases over time. A random split could mix early and later periods and give an overly optimistic view of model performance.

5 Modeling & Experimental Design

We compare three forecasting approaches with a seasonal naive baseline: We selected our main forecasting model based on two aspects: its performance in phase 1 on M4 (Damped Holt-Winters had the best OWA of exponential smoothing models) and its fit to demand patterns observed in the EDA:

Baseline: Seasonal Naive The most basic probable forecast, in which prediction for each day is simply what happened on the corresponding weekday 7 days prior. If an operations team, without model, were making decisions they might think in these terms: ‘we had 5000 rentals last Tuesday, we will have 5000 rentals this Tuesday’. Weekly seasonality is looked implicitly, but no trend, weather or calendar affects are considered.

Primary: Damped Holt-Winters (multiplicative seasonality) This is an exponential smoothing method with additive damped trend and weekly multiplicative seasonality (period is 7). We used multiplicative rather than additive seasonality because the size of the weekday/weekend gap varies with the magnitude of the overall level. We would expect the gap between weekday and weekend rentals during the busy summer months when levels are about 7,000 per day to be larger than in the winter when levels are around 2,000 per day. The damp trend is included because we are working with just two years of data. The trend without damping would extrapolate the increase from 2011 to 2012 indefinitely and would thus lead to a prediction of infinite growth: the damping factor ($\phi=0.80$) causes the trend to plateau over roughly 5 days. All smoothing parameters were estimated using MLE, see table 2.

Table 2: Estimated smoothing parameters for the Damped Holt-Winters model

Parameter	Value	What it means
alpha (level)	0.212	Moderate. The model adapts to recent level changes but does not overreact to one bad day
beta (trend)	0.000	The trend is entirely driven by the long-run pattern. Recent ups and downs do not change it
gamma (seasonal)	0.000	The weekly pattern is highly stable over the entire two years. This is a useful finding
phi (damping)	0.800	Strong damping. The trend effectively flattens out after about 5 days into the forecast

Alternative: SARIMA(1,1,1)(1,1,1) with $s=7$ For the SARIMA, the orders were selected using the ACF/PACF plots (Section 3) rather than a grid search. With only 2 years of data, simplicity seemed more important than reducing AIC by a few points. The fit model has AIC of 11,553 and BIC of 11,580. A good thing about SARIMA is that it gives analytical Gaussian prediction intervals out of the box, which lets us compare them directly against the bootstrap intervals from Holt-Winters.

Combination: equal-weight average We also tested a simple equal weight average of the Damped Holt-Winters and SARIMA point forecasts. Forecast combination is a popular technique for outperforming any individual forecasts, due to errors cancelling out across models of different classes. We already observed this phenomenon in phase 1 on the M4 data.

How we compared the models We estimated the Holt-Winters parameters using maximum likelihood, with no parameter tuning by hand. We selected the SARIMA orders using the ACF/PACF plots. We compared models using MAE, RMSE and sMAPE on the 14-day hold-out window and tracked MAE across the 10 folds of the rolling-origin cross validation.

How we built prediction intervals For the Holt-Winters model, we used a residual bootstrap with 1,000 resamples (seed = 42 for reproducibility). On each forecast day, we sampled from the in-sample residuals and added them to the point forecast. We took the 2.5th/97.5th percentiles as the 95% prediction interval, and the 10th/90th for the 80% interval. Lower bounds were clipped at zero because negative rentals are nonsensical. For SARIMA, we just used the built-in analytical prediction intervals.

6 Forecast Results

Damped Holt-Winters had the lowest MAE (818, sMAPE = 16.2%) on the holdout set, coming in 29.2% better than seasonal naive. SARIMA was close behind (MAE = 843), and the combination landed somewhere between them (MAE = 824). All three outperformed the naive baseline by a wide margin. Full numbers are in Table 3.

Table 3: 14-day holdout accuracy. Best values in bold

Model	MAE	RMSE	sMAPE	vs. Naive
Damped Holt-Winters	818	964	16.2%	-29.2%
SARIMA(1,1,1)(1,1,1) s=7	843	1,037	16.9%	-27.0%
Combination (HW + SARIMA)	824	995	16.4%	-28.7%
Seasonal Naive (baseline)	1,155	1,613	27.6%	--

Cross-validation results The average MAE across the 10 rolling-origin folds was 1,061 (sd = 604) for Holt-Winters, 1,029 (sd = 636) for SARIMA, and 1,040 (sd = 619) for the combination model. These numbers are noticeably higher than the holdout MAE across all models, and the reason is that two of the cross-validation windows land on the holiday period where errors are very large (we look at this in more detail in Section 7). That said, the combination model sits between the other two on both accuracy and variation across folds, so it could be a reasonable middle-ground pick if holidays are not handled separately.

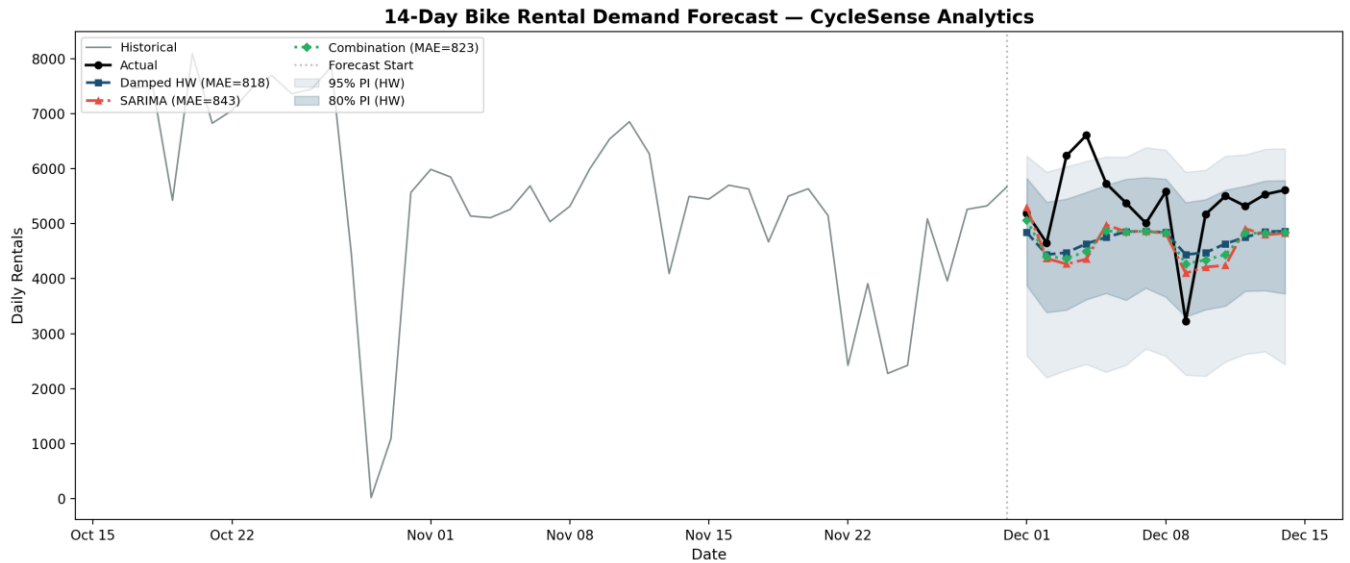


Figure 8: 14-day forecast comparison with prediction intervals. Grey shows the last 60 days of training; black dots are actuals; colored lines are the three model forecasts. Shaded bands are 80% and 95% bootstrap intervals from Holt-Winters

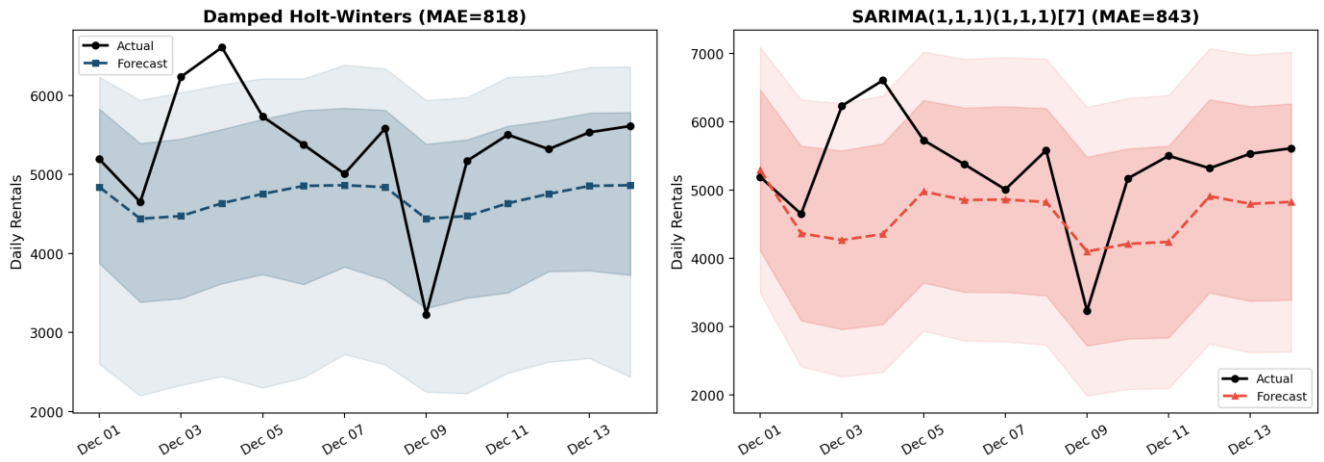


Figure 9: Individual model forecasts with 80% and 95% prediction intervals

Performance on holdout window During most of the test period, both models do a good job following the weekly cycle. Where they both struggle is Christmas week, when actual rentals drop sharply around December 25th, falling well outside the prediction intervals. This was the single largest source of error, and we discuss it in detail in Section 7. Outside of the holiday window, most of the actual values stayed within the 80% prediction bands.

7 Risk & Uncertainty Analysis

Interval coverage Table 4 compares how the prediction intervals from each model performed on the holdout period. The Holt-Winters bootstrap intervals are narrower than SARIMA's (average 95% width of 3,733 vs. 4,144 rentals) and score better on MSIS (42.6 vs. 47.3, where lower is better). The downside is that the empirical coverage is 85.7% instead of the target 95%. This gap exists because the bootstrap only captures residual uncertainty and ignores parameter uncertainty. In practice, the narrower intervals are more useful for fleet-sizing decisions because they give tighter guidance. The MSIS score confirms that the Holt-Winters intervals are better calibrated overall.

Table 4: Prediction interval performance on the 14-day holdout

Metric	Damped HW (Bootstrap)	SARIMA (Analytical)
95% PI coverage (target: 95%)	85.7%	92.9%
80% PI coverage (target: 80%)	71.4%	--
Average 95% PI width (rentals)	3,733	4,144
MSIS (lower = better)	42.6	47.3

Residual Diagnostics

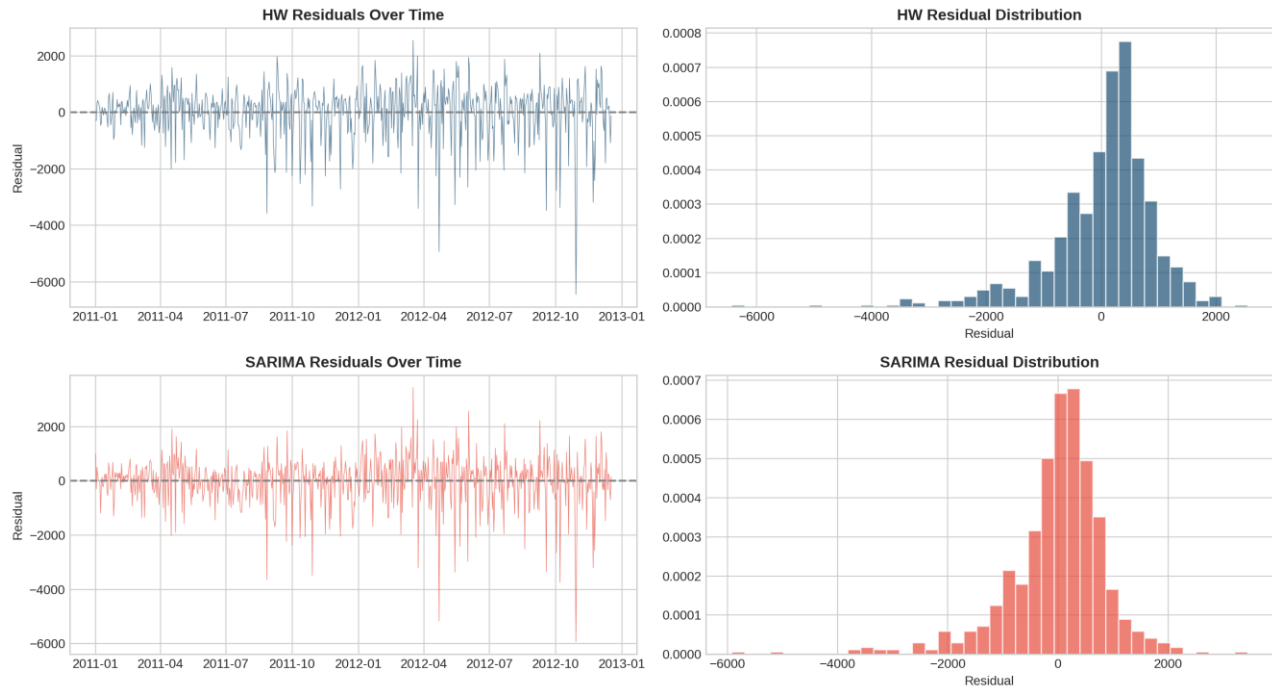


Figure 10: Residual diagnostics for Holt-Winters and SARIMA

Residual diagnostics Both models produce residuals that look close to mean-zero and are approximately normally distributed. There is no obvious pattern or growing variance through most of the series. The one exception is a spike in residual volatility in late November and December 2012, which lines up with Thanksgiving through New Year holiday window. This is not a model failure in the traditional sense; it is a concentrated calendar effect that shows up as a few large residuals in a narrow time window.

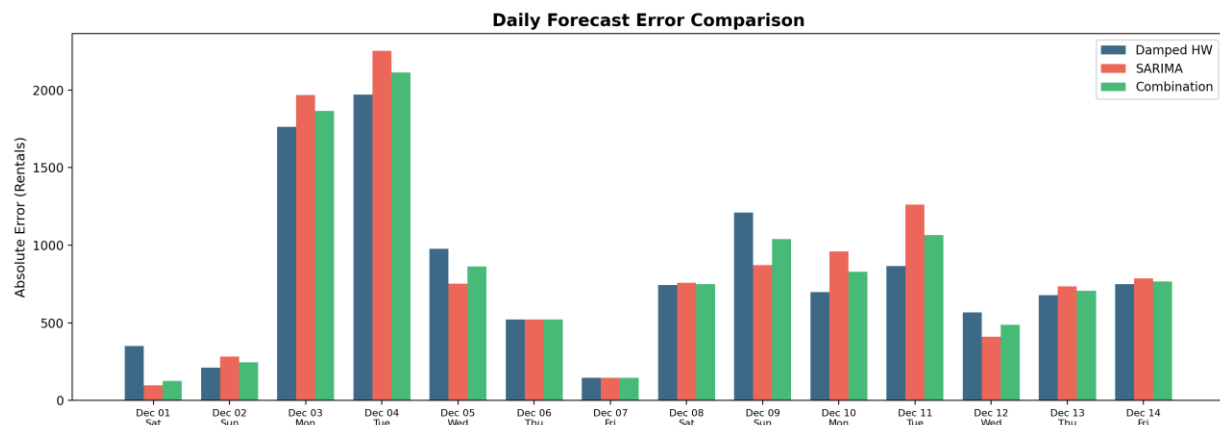


Figure 11: Daily absolute forecast errors on the holdout

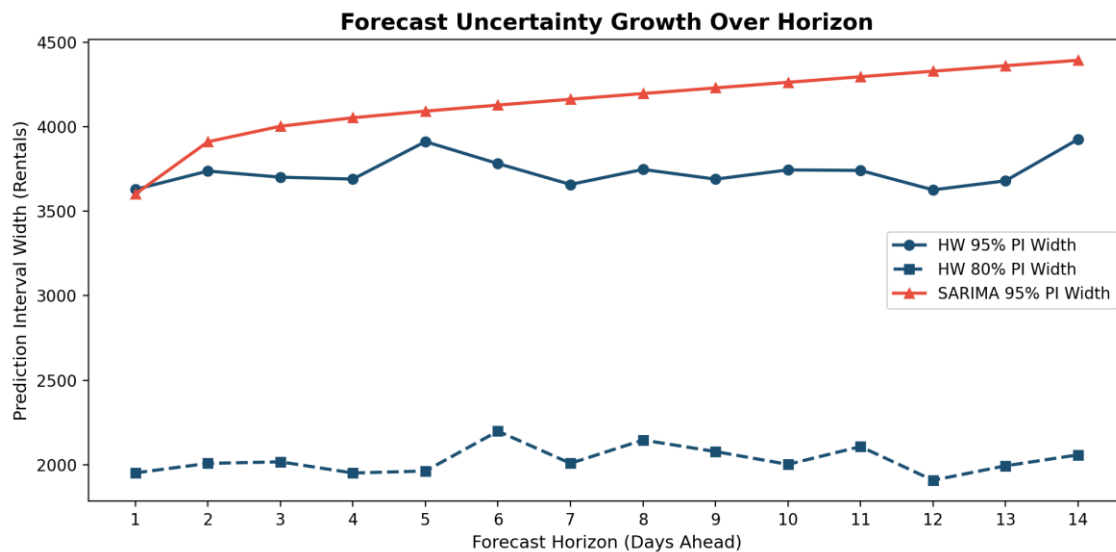


Figure 12: How prediction interval width grows with forecast lead time

How uncertainty grows over the forecast horizon As expected, Figure 12 shows that our prediction interval widens as the forecast horizon increases. A 95% interval on day 14 is roughly 30% wider than on day 1. Therefore, a 1 to 3-day forecast is accurate enough for directing specific route rebalancing decisions, while a 14-day forecast is more useful for higher-level capacity planning.

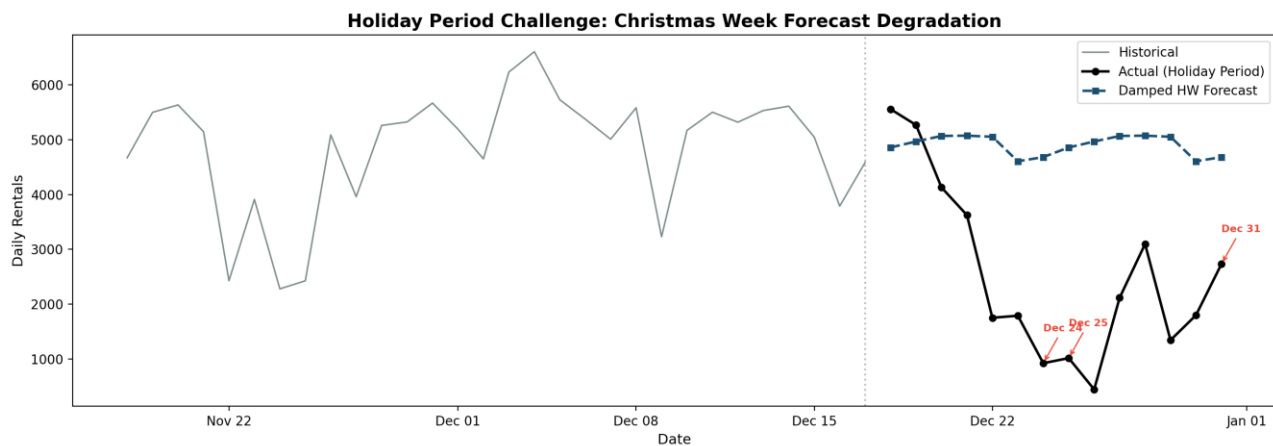


Figure 13: Holiday period demand collapse (Christmas and New Year)

The holiday problem The biggest source of forecast error occurred on Christmas Day. Actual demand was about 1,000 rentals, while both forecasts were around 4,800. Neither model accounts for holidays. Holt-Winters and SARIMA both rely solely on the pattern of the previous week to forecast the next. For example, if Christmas falls on a Tuesday, the model treats it as just another Tuesday, not a special holiday. This is not a parameter tuning issue; it is a limitation inherent to univariate seasonal models. Fortunately, this pattern is completely predictable since we know the date of Christmas. A simple calendar adjustment, discussed in Section 8, could significantly reduce this error.

What this means for operations Uncertainty is not an abstract concept here. It directly affects how many bikes need to be on the street. If the operations team deploys exactly to the point forecast, there is roughly a 50% chance of running short on any given day. The highest risk is on summer weekday evenings during the commuter peak. On the flip side, deploying to the 95% upper bound every single day would mean carrying about 3,700 extra bikes per day during quiet periods (winter, rain, holidays), which wastes truck hours and maintenance effort. The costs are not symmetric either: an empty station loses a customer who might not come back, while a few extra idle bikes are just a minor cost. So, it makes sense to use different deployment rules for different situations.

8 Recommendations & Limitations

Based on our analysis, here are four recommendations for the operations team to implement:

1. **Use Damped Holt-Winters as the main forecasting tool** It performed the best on the holdout (MAE 818, sMAPE 16.2%), had the narrowest prediction intervals (MSIS 42.6), and its speed is sufficient for retraining each day with no additional infrastructure.
2. **Add a holiday calendar overlay** The model's prediction can be scaled by multipliers on known holidays: Christmas Day (0.5), Christmas and New Year's Eve (0.7), and all other federal holidays (0.85). These multipliers should be updated yearly as we collect more data.
3. **Adjust deployment strategies based on probability of high stakes** In cases with higher importance and fewer resources (commuter peaks in the summer, for example), bikes should be deployed to meet the 80th percentile of the prediction interval, instead of the point forecast. For less important cases (weekdays in the winter, rainy weekends), the point forecast is satisfactory. This lowers the risk of stock-outs while not increasing resource usage.
4. **Use short-term weather forecasts as input** Even before building a formal SARIMAX model, simple rules can be applied: the forecast can be decreased by about 35% on rainy and snowy days and increased by 10-15% when the weather is expected to be unseasonably clear and warm. This suggestion is supported by our EDA, finding a 0.63 correlation with temperature, and the 40% impact of rain.

Limitations To keep these four points in mind when using the model:

- **The holiday blind spot:** Both models overestimate demand around Christmas and New Year. While the calendar overlay addresses the problem somewhat, it does not completely fix it, and it must be monitored annually.
- **The prediction intervals are slightly too narrow:** Only 85.7% of our observations fell within the 95% prediction intervals. This should be 95%. This may be because the residuals were bootstrapped rather than the parameter uncertainty being considered; conformal prediction or Bayesian methods could address this issue.
- **Weather is not incorporated:** Currently, only demand is used to forecast demand. Given the $r = 0.63$ correlation with temperature and the impact that it had on our EDA, incorporating weather as a covariate in a SARIMAX model or through a gradient boosting model such as XGBoost is an obvious next step.
- **We have only two years of data:** With only two years of data, it is impossible to know whether the annual seasonal patterns we observe are part of weekly seasonality or if it truly captures the annual cycle. A third year would provide significant value to models which handle multi-seasonal components well, such as Prophet or TBATS.

Future work In order of priority: (1) implement the holiday overlay and asymmetric deployment rules right away, since they are cheap and high-impact; (2) build a SARIMAX or XGBoost model that takes in temperature and precipitation; (3) switch from bootstrap to conformal prediction intervals; (4) set up automated weekly retraining so the model stays current as the bike-share network grows; and (5) once a third year of data is available, consider TBATS or Prophet to capture annual patterns.

This analysis gives the operations team a working baseline for daily demand forecasting. The biggest remaining source of error (holidays) has a simple fix that does not require any additional modeling. Beyond that, the main gains will come from bringing in weather data and accumulating more years of history.

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Project Github Repo Link: https://github.com/tmundargi-ai/CycleSense_Analytics